Measurement and Analysis of Tissue Temperature During Microwave Liver Ablation

Deshan Yang, Mark C. Converse*, *Member, IEEE*, David M. Mahvi, and John G. Webster, *Life Fellow, IEEE*

***Abstract*—We measured tissue temperature changes during *ex vivo* microwave ablation (MWA) procedures for bovine liver tissue. Tissue temperature increased rapidly at the beginning of the MW power application. It came to a plateau at 100 °C to 104 °C before it increased again. We split the changes of tissue temperature versus time into four phases. This suggests that tissue temperature changes may be directly related to tissue water related phenomena during MWA, including evaporation, diffusion, condensation and tissue water composition. An additional analysis indicated the lesion boundary at $\sim$ 50 °C to 60 °C temperature. We also measured the water content of ablated tissue lesions and examined the relationship of tissue water content and tissue temperature by mapping temperature to remaining tissue water after ablation. The results demonstrate significant tissue water content changes and lead to a better understanding of tissue water movement.**

***Index Terms*—Microwave liver ablation, temperature monitoring, water content.**

I. Introduction

MICROWAVE ablation (MWA) is a promising technology for the treatment of hepatic tumors. The goal of MWA is to destroy the tumor along with a 1 cm margin of normal hepatic tissue. This technology has been used in both intra-operative and percutaneous approaches for primary hepatocellular carcinoma and hepatic metastasis of colorectal carcinoma [1]–[3].

Much of the previous research into MWA has focused on the design and optimization of the antenna applicator utilizing predominantly electromagnetic (EM) computer simulations and experimental measurement of EM fields and final lesion size. Less prevalent in the literature are temperature measurements and the use of thermal models to predict antenna performance or tissue response. This may be due to the complexity of heating mechanisms and difficulties with measuring tissue temperature and other relevant variables during MWA procedures. While more effort has been put into thermal modeling of radio frequency ablation (RFA), the heating mechanisms of MWA are different from RFA and are expected to be more complex due to higher temperatures seen with MWA. Preliminary studies have shown that MWA was able to heat tissue to temperatures higher than 125 °C, much higher temperatures than RFA. As such, during the course of heating, tissue will lose water content due to the generation of water vapor or by the diffusion of liquid water from the treated cells. If a significant amount of evaporation takes place, gas pressure will increase and the vapor will diffuse into lower pressure areas where the temperature is lower. One would expect this water vapor to condense in this lower temperature region adding energy and increasing water content. Some tissues, thus, gain water content and heat energy during the condensation process while others lose water content and heat energy during the evaporation process. The entire process of water evaporation, water vapor diffusion and condensation is a process of water movement and energy movement and may be as significant as direct thermal conduction [4]–[6].

As well as adding a new heat transfer mechanism, the movement of water and specifically the loss of water in some volumes of tissue are expected to affect other tissue properties. Both tissue dielectric properties and thermal properties are directly related to tissue water content [7]–[9]. Changes in tissue thermal properties will directly affect the heat conduction within tissue and changes in tissue dielectric properties are expected to lead to changes in the location of energy deposition within the liver.

If the water-based phenomena that occur during microwave ablation (MWA) were better understood, incorporation of these effects into EM/thermal models could lead to more accurate models of MWA.

We have found no existing study that provides quantitative measurement or analysis of tissue water content and associated temperature profiles during the course of MWA of tissue. We present results of temperature measurements during MWA and attempt to explain gross tissue changes based on temperature profiles measurements of water content. To examine how the water content changes, we measured the water content distribution in ablated tissue. Our initial results confirmed our expectation that there are significant changes in tissue water content during MWA.

II. Methods

All experiments were performed *ex vivo* with bovine liver obtained from a local slaughter house. Liver tissue was refrigerated overnight and the initial temperature at the beginning of the experiments was approximately 6 °C–9 °C. Homogeneous blocks of bovine liver, at least $8 \times 8 \times 5$ cm, were used for the experiments. Initial water content was 73% by weight, measured with a wet—dry procedure on larger tissue pieces to reduce measurement errors. The coaxial antenna was inserted into the tissue and connected to a MW generator through a 1-m-long flexible

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coaxial cable. A lesion was created after 55 W of microwave power was applied to the liver tissue for a controlled duration.

A. Temperature Monitoring

We used a Luxtron Fluoroptic thermometer model 3100 to measure tissue temperature during MWA of *ex vivo* bovine liver tissue. A fluoroptic thermometer was selected because its fiberoptic temperature sensors are unaffected by microwave radiation and have minimal affects on the specific absorption rate (SAR) pattern of the antenna. The antenna used in this study is a floating sleeve antenna we have previously described [10]. The antenna was designed to work in liver and the design was based on properties from Gabriel *et al.* [11]. The reflection coefficient is approximately -17 dB which created a good initial match to the tissue.

Fig. 1(a) shows the placement of the microwave coaxial antenna and fiberoptic temperature sensors. The antenna was inserted 7 cm deep into the tissue, referenced to the antenna tip. In addition to the antenna, fiberoptic sensors were inserted parallel to the antenna, into the liver tissue guided by 14 gauge (2.1 mm diameter) biopsy needles through holes in a 1 cm thick plastic template. This ensured that the antenna and the needles were placed parallel to each other in the tissue. The needles were 15 cm long, with a needle guide and a 15-gauge (1.83-mm-diameter) stylet. The needle guide and the stylets were pushed together through the liver block, and then the stylets were withdrawn. Temperature sensors, each paired with a section of thin plastic fiber (0.5 mm in diameter), 12 mm long, were introduced into the needle guides. The temperature sensor fibers were inserted to a position 5 mm proximal to the antenna slot. This is the axial location of maximum power deposition as predicted by an electromagnetic simulation of the antenna in liver. This maximum lesion radius was also confirmed in the measurement of an actual lesion, shown in Fig. 5(a). Radial positions for the sensors were chosen based on practical considerations and based upon recognizing that there will be greater temperature changes closer to the antenna. The plastic fibers were inserted 2 cm beyond the temperature sensor fibers. The needle guides were then withdrawn from the tissue while the fiberoptic temperature sensors and the plastic fibers left in the liver tissue.

Each temperature sensor was connected to a Luxtron 3100 thermometer, which was connected to a PC via its serial port for data collection. Temperature values from all sensors were updated by the thermometer four times a second and were recorded by the PC for the entire ablation procedure. After ablation, the temperature sensors and the antenna were removed, leaving the plastic fiber sections behind to mark the temperature sensor locations. After slicing the tissue into transverse segments, plastic fiber locations were measured. Temperature sensors were assumed to be at the same positions as the plastic fibers.

B. Water Content Measurement

After ablation the water content measurement proceeded as follows: the antenna was smoothly drawn out, the liver tissue was cut into 2 pieces along the trace of the antenna insertion. A 5-mm-thick slice was cut from one of the two pieces, at a distance 5 mm from the slot, perpendicular to the antenna insertion direction, as shown in Fig. 1(a). A strip of tissue was then cut from the slice and cut into 1-mm-thick pieces, as shown in Fig. 1(b). The plane 5 mm above the slot was chosen as this is the position along the axis of the antenna with the largest lesion radius, based upon simulation and experimental observation.

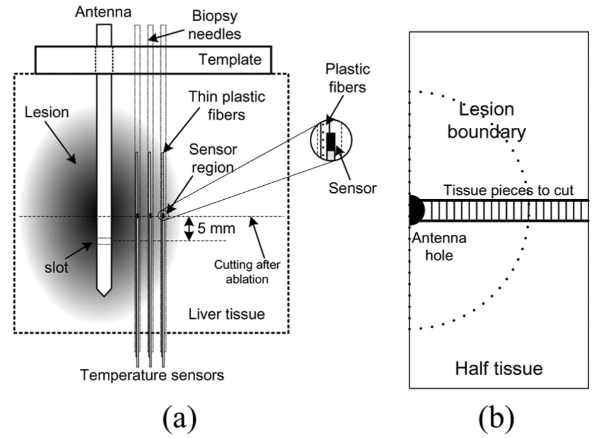


Fig. 1. (a) Schematic of the *ex vivo* experimental setup. The fiberoptic temperature sensors are placed 5 mm away from the antenna slot in the longitudinal direction, and various distances away from the antenna body in the radial direction. The master slice 5 mm thick was cut at 5 mm above the antenna slot. (b) Illustration of the distribution of 1-mm-thick samples cut from the master slice.

Weights of all tissue pieces were measured with an Ohaus Explorer E11140 digital analytical balance with 0.1-mg resolution. Tissue pieces were vacuum-dried overnight with an Edwards ETD4 Tissue Dryer, and weighed the next morning. All measurements were performed as quickly as possible to minimize the effects of evaporation from the surfaces of the tissue.

The mass fraction of tissue water remaining after ablation, w_r, is calculated as

$$w_r = \frac{M_w - M_d}{M_B} = \frac{0.27\,(M_w - M_d)}{M_d} \tag{1}$$

where, M_w is the measured tissue weight before drying, M_d is the measured dry tissue weight, and M_B is the weight before ablation. M_B is calculated as

$$M_B = \frac{M_d}{0.27}. \tag{2}$$

Equation (2) indicates the water content of normal pre-ablated liver is 73%. This is based upon measurements of large pieces of tissue where water loss due to cutting of tissue and evaporation from the surface are negligible. This value is consistent with literature [9].

A note regarding measurement errors: results of the water content measurement above 50% mass fraction are not presented due to water drip from cut tissue. During the cutting stage of the experiment we noticed a significant loss of water from the higher water content tissues pieces. This led to a discrepancy between the measured values of the high water content (normal) tissue outside the lesion (based on small pieces) and our normal liver water content measurements taken for calibration purposes based on large pieces. Water loss was

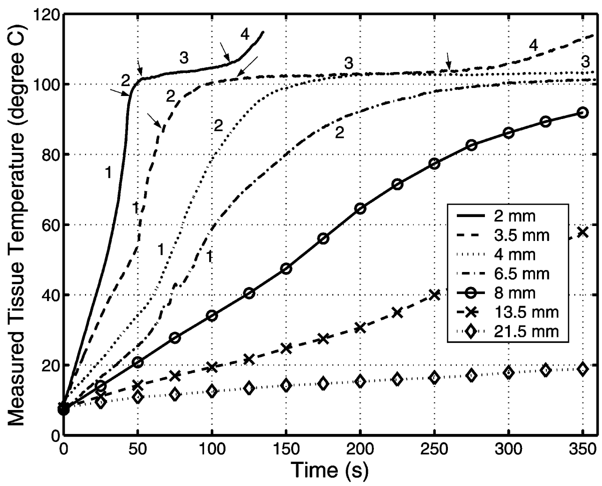


Fig. 2. Plots of measured temperature versus time. The MWA procedure was performed with 55 W for 360 s. Sensors were 5 mm above the antenna slot longitudinally. Temperature curves are labeled by the distance of the temperature sensors to the antenna surface, measured after heating.

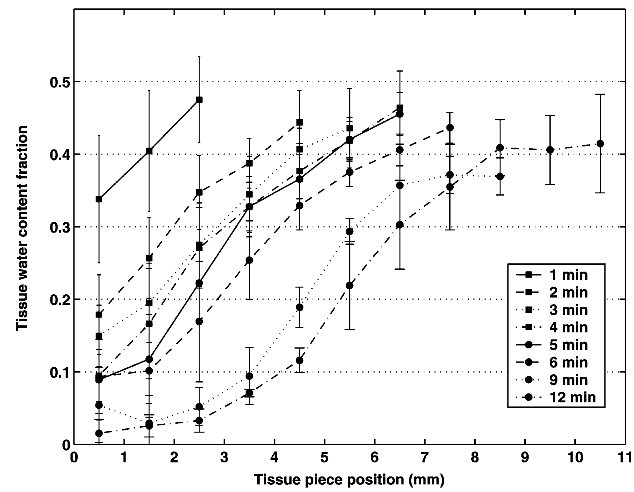


Fig. 3. Plots of post ablation water content mass fraction w_r, calculated according to (1). Mean results from multiple measurements for the same duration are plotted as solid or dashed lines. The vertical bars represent the standard errors at the corresponding data points. The data points are plotted at the position where the center of each tissue piece is located on the lesion strip. Position value is the distance from the edge of the antenna insertion hole to the center of the tissue piece. 0 is at the edge or the surface of the antenna when the antenna is in the tissue.

only observed in tissue pieces with higher than 50% of water content, therefore measurements on such tissue pieces were unreliable and are not presented. For tissue pieces of lower water content, below 0.5, no visible water loss was detected, nor was expected, so the measurement results are considered acceptable. For the purposes of understanding water related phenomena near the antenna these are clearly the most significant.

Cutting and weighing of the tissue after ablation took approximately 1 to 3 min. During this delay, tissue water may have diffused from a high water content area to low water content area—the highly ablated region. The overall effect would be to artificially increase the tissue water content near the antenna body, and artificially decrease the water content near the lesion boundary. The estimated water diffusion constant in normal tissue is approximately 10^{-9} to 10^{-10} m^2/s [12]. Diffusion rates may be even higher in the highly ablated area because the tissue is porous. Nevertheless our data do not suggest that the amount of diffusion was quantitatively significant and initial computer simulations utilizing the base water diffusion constant support this hypothesis.

III. Results

Numerous samples were ablated with temperature measurements at various radial locations. Taking into account inhomogenities in the liver, all measurements showed the same curve shapes and range of temperatures. Fig. 2 shows temperatures from some representative measurements during ablation for different radial locations. The shape of the curves in Fig. 2 show four distinct regions, labeled 1, 2, 3, 4 on the graph. What separate the regions are the temperature ranges and curvature. Region 1 of the temperature curve is slightly concave, starting at the initial tissue temperature and ending between 70 °C and 90 °C. Region 2 is convex and appears to be a transition region between regions 1 and 3. In region 3 the temperature curve becomes almost flat and slowly increases from 100 °C to 104 °C. After the temperature reaches 104 °C, temperature increases quickly again in region 4. Most curves showed these different regions although changes are more gradual as distance from the antenna increases.

Fig. 3 shows the results of water content measurements on lesions created by MWA with durations of 1, 2, 3, 4, 5, 6, 9 and 12 min where w_r was calculated for each case. We performed at least four measurements for each duration. Fig. 3 shows the mean results and standard deviations. We used the analysis of covariance (ANCOVA) statistical model to analyze the measured data and show statistical significance. The results show that tissue water content changes significantly $(p < 0.001)$ with ablation duration by controlling the tissue piece position. Tissue water content also changes significantly $(p < 0.001)$ with tissue piece position by controlling the ablation duration. The statistical results are consistent with the plots.

To better correlate tissue temperature with water content, Fig. 4 plots tissue temperature and water content as a function of time at different locations. In Fig. 4, values of tissue water content are the ratios of the remaining tissue water mass, after ablation, to the normal tissue mass before ablation. According to bulk measurements, normal liver tissue has about 73% water content. Fig. 4 indicates that, for all locations shown, as tissue temperature increases to 100 °C, water content drops to 45%–50%. For locations near the antenna (2, 3, and 3.5 mm), as the temperature reaches 104 °C, water content drops to approximately 30%–35%. Note that for locations 4 and 5 mm, although the temperature does not reach 104 °C, water content at those locations eventually drops to 35%. Based upon the water content of normal liver, it appears as though approximately 30% of the water in the tissue is lost by approximately 100 °C, 30% is lost during the transition in temperature from 100 °C to 104 °C, and the remaining 40% is lost as the temperature continues to increase above 104 °C.

As part of the measurement of tissue temperature during ablation we measured the temperature at the lesion edge. Fig. 5

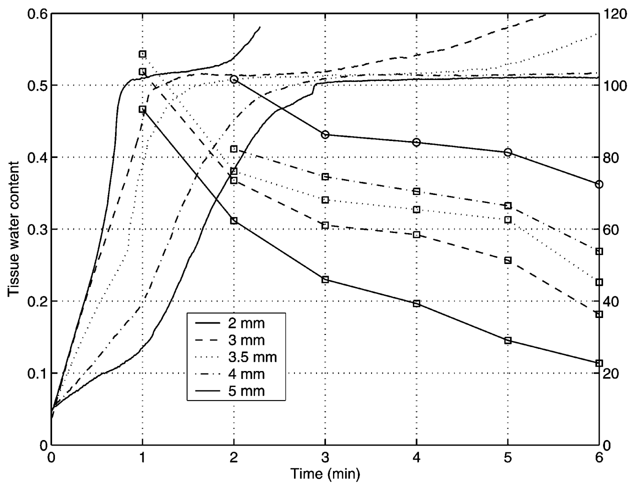


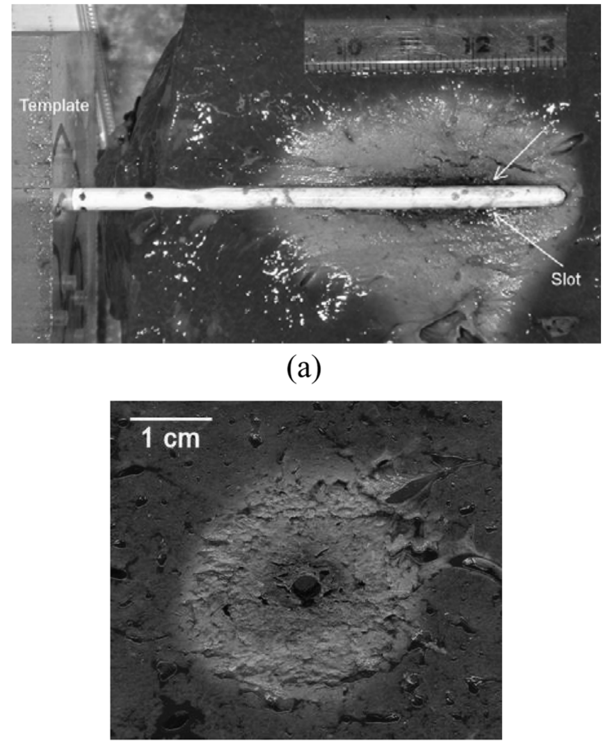
Fig. 4. Tissue water fraction and measured temperature versus ablation duration. Curves are labeled by the distance of the temperature sensors to the antenna surface. Curves without markers correspond to plots of tissue temperature. Curves with markers are plots of tissue water content with the markers being data points.

shows a picture of ablated tissue. The lesion is identified as the region in which tissue color changes from dark red color to light red color. In Fig. 5, this is the transition from the darker to lighter region. The antenna hole is approximately 3.5 mm in diameter. Based upon 9 experiments, the length from lesion boundary to the edge of antenna hole had a mean value of 14.5 mm and standard deviation of 0.9 mm. The mean lesion diameter was 32.5 mm. Fig. 3 shows that tissue temperature was 60 °C at 13.5 mm after 6 min ablation. This suggests that *ex vivo* liver lesion boundary had a temperature of 55 °C–60 °C. This is very similar to the 50 °C, determined in [13] for RF ablation.

IV. Discussion

The results suggest that MWA has several effects on tissue water properties. First, tissue is desiccated to a greater extent near the antenna. This is expected both from an observation of the lesion and from a basic understanding of the SAR pattern (i.e., where energy is deposited) and heat transfer. Second, even after only 1 min there is significant water loss in the tissue up to 3 mm away from the edge of the antenna. This illustrates the volume heating effect we expect from MWA. Microwaves emitted from the antenna propagate through the tissue, resistively heating the tissue as the wave attenuates. This leads to a direct deposition of energy in a larger volume of tissue than seen during RF ablation. Third, the change in water content seen early and over a significant volume can be expected to affect both the thermal and electromagnetic properties. This may affect them enough to significantly alter the SAR pattern and/or the dynamics of heat flow that occurs in conjunction with the direct deposition of energy due to the microwaves. Finally, water content drops to near 0% for tissue very close to the antenna for durations of greater than 6 min. This indicates charring of the tissue near the antenna.

We considered the relationship between temperature and water content. For points closer to the antenna we see a slight increase in slope of the temperature plot from 30 to 90 s. Comparing plots of temperature at different positions in Fig. 2, we



(a)

(b)

Fig. 5. Ablated bovine liver tissue. (a) Longitudinal section, the maximum lesion radius is located at about 5 mm from the slot along the longitudinal direction. (b) Cross section. Both lesions were created separately by applying 55 W for 360 s. Lesion diameters are approximately 31 to 33 mm in diameter.

see that at the time when the slope changes, points closer to the antenna have reached temperatures approaching 100 °C. Fig. 4 shows significant loss of water in those regions. This water is likely moving outward radially and could be contributing to radial heat transfer, due to diffusion of high temperature liquid water or vapor. Another possibility is that as the water moves outward, the dielectric and thermal properties of the tissue close to the antenna change. This would affect thermal conductivity or increase the direct deposition of energy to the lower temperature regions as less energy is absorbed in the higher temperature regions due to loss of tissue water.

As the curve approaches 90 °C–100 °C, the rate of temperature change decreases. Fig. 4 shows that water content begins decreasing. Either water is diffusing away from this region or likely some evaporation is beginning to occur. Ultrasound measurement during ablation has shown anomalies attributed to air bubbles [14], [15]. As energy is required to convert liquid water to vapor, less is available to increase the temperature. The temperature curve eventually flattens out and slowly increases from 100 °C to 104 °C, in conjunction with this plateau the water content begins to drop significantly. This lends credence to the idea of water evaporation. Note, due to the complex structure of tissue and the way that water is distributed, as well as what is likely a relatively slow diffusion rate of water vapor from the region, we would not expect the temperature to remain at a "boiling" temperature until all water is evaporated.

After the plateau at approximately 100 °C–104 °C, the temperature again begins to rise. At this point the mass fraction of

$$W(T) = \begin{cases} 0.778 - 0.778 \times \exp\left(\frac{T-106}{3.42}\right) & T \leq 103 \\ 0.0289T^3 - 8.924T^2 + 919.6 \times T - 31573 & 103 < T \leq 104 \\ 0.778 \times \exp\left(\frac{T-80}{34.37}\right) & T > 104 \end{cases} \quad (3)$$

water has dropped to approximately 30%–35% of the original tissue mass or about 40% of the initial water mass. This may be intracellular water or water that is somehow bound to the tissue and requires significantly more energy to evaporate. Temperature continues to rise until it is beyond the range of our measurement tools.

We expect tissue water content to play a significant role in MWA. To include the effects of water evaporation and movement into a thermal model of MWA we have attempted to create a function that correlates tissue temperature to water content.

Ramachandran *et al.* measured water evaporation from heated tissue [16]. For liver tissue, their results showed that tissue water evaporation started when tissue temperature reached 70 °C and approximately half of the tissue water content was lost by the time the temperature reached 104 °C [16]. Haemmerich *et al.* introduced a new method to measure specific heat, which reduced the effects of surface evaporation [17]. This method utilized larger liver tissue pieces and is more consistent with the environment seen by the liver during MWA. Their results indicated that the liver tissue began losing water at 80 °C. The results also showed that there is a 17% increase in tissue specific heat by 83.5 °C and 15% tissue mass reduction by 92 °C. In [18], we introduced an effective specific heat, in which the latent heat of evaporation is incorporated into a temperature-dependent specific heat. The change in specific heat at high temperature that Haemmerich *et al.* observed is consistent with our effective specific heat, which could imply that this change is due to water vaporization. Their observation of 15% tissue mass reduction seems too high. We think such mass reduction was mainly caused by loss of tissue water from the blood vessels under the gas pressure of water vapor, rather than caused by tissue water evaporation.

As the method proposed by Haemmerich *et al.* is more consistent with the environment seen by the liver during ablation, we incorporated his conclusions and our experimental measurements into a water content function based upon tissue temperature, as in (3) at the top of the page, where T is tissue temperature [°C] and W is ratio of remaining tissue water mass to the mass of normal tissue before heating.

Fig. 6 shows the proposed functional relationship between tissue temperature and tissue water evaporation based upon our measurements of water content, as well as results/observations by Ramachandran *et al.* and Haemmerich *et al.* These additional sources of information on the relationship lead to the difference between our measured data and the proposed functional relationship. Equation (3) was fitted to data for temperature from 80 °C to 104 °C by an exponential function. For temperatures less than 80 °C, tissue water content is 73%, which is the value we measured for normal liver. At 83.5 °C, the effective specific heat [18] is increased by about 17% over regular effective heat.

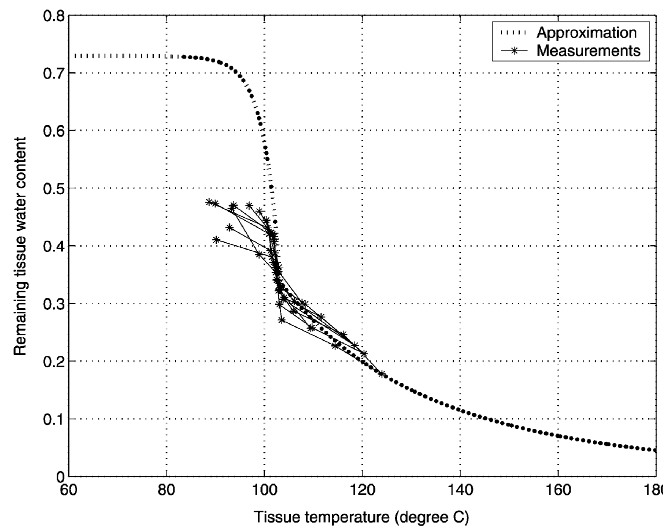


Fig. 6. Remaining mass of tissue water versus tissue temperature after ablation. Tissue water and tissue temperature from matched ablation durations and positions in lesion are plotted as thin solid lines with asterisks. Only data points lower than 0.5 were used to create the approximation line because data points higher than 0.5 were not reliable. The dotted line shows the mathematical approximation. Normal tissue has about 73% water content in mass.

For temperature over 80 °C the water content begins to drop at an increasing rate to about one-half by 104 °C, and then exponentially decays as the temperature get higher. This exponential decay is relatively arbitrary as we have few data above 120 °C and have found nothing in the literature. In the equation, both W and its first-order derivative are continuous.

V. Conclusion

This study presents results of tissue temperature and tissue water content measurement during MWA. Our initial results indicate a relationship between tissue temperature and tissue water content. Based on this perceived relationship, a function to related water content to tissue temperature is presented. Although performed for a specific set of conditions and one antenna geometry, useful insights were obtained and may be applicable to a broader range of conditions.

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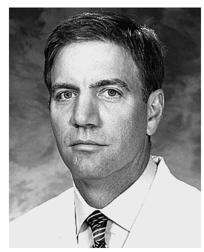

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He is the editor of the most-used text in biomedical engineering: *Medical instrumentation: application and design, Third Edition* (Wiley, 1998), and has developed 22 other books including the *Encyclopedia of medical devices and instrumentation*, 2nd ed. (Wiley, 2006) and 180 research papers.

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